Team 17: Six Pack

CENTRALITY WITH HERDING IN SOCIAL NETWORKS

References

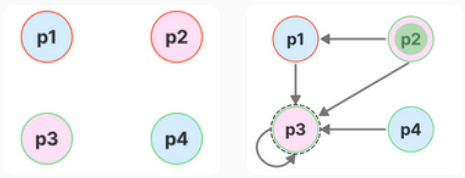
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Experiment Goal:

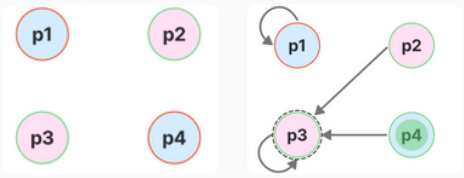
Our project aims to empirically study opinion dynamics in small groups. Specifically, we wanted to see whether a group discussion can change or reaffirm people's beliefs. We hoped to understand how susceptible people are to changing their stance when exposed to opposing viewpoints.

Pre and Post discussion Graphs:

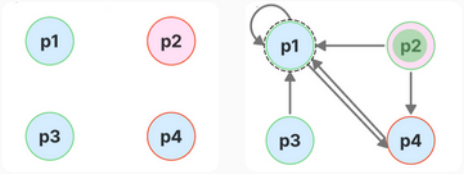
Group 1:



Group 2:



Group 3:



colors-> gender ; **border**-> green (yes reservation) red (no reservation) ; **edges**-> who they thought was the most influential/ spoke the most ; **dotted**-> who was the most central ; **green transparent sphere** -> who changed their opinion

Our Approach:

We designed the experiment with the following structure:

1)Pre-discussion Form (Google Form #1): Participants first filled out a form stating their initial stance on gender-based reservation.

2)Group Discussion (Minimum 10 minutes): Three groups of four participants each discussed for at least 10 minutes, guided by a moderator who asked provocative questions to ensure a realistic environment that allowed influence, persuasion, and conflict to emerge.

3)Post-discussion Form (Google Form #2): A second form captured their final stance, opinion change, and who they found most influential.

Results:

Across three group discussions, both starting with equal diversity of opinion (2-2), we found:

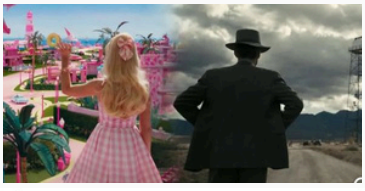
- Opinion Change: In every group, at least one participant shifted their stance, moving the split from 2–2 to a 3–1 majority.
- Majority Reinforcement: Participants whose stance stayed the same and aligned with the final majority felt reassured in their stance.
- Reasons for Change or Resistance: Those who changed their opinions cited new perspectives and strong arguments from one key participant, while minority participants wrote very little, suggesting low engagement or stubborn beliefs.
- Influence & Centrality: The person judged "most influential" was also the most central, and their stance consistently became the group's final majority position.

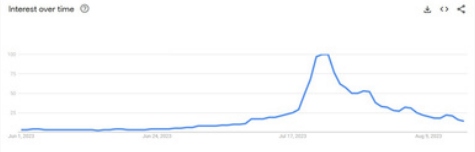
Research Idea:

To study how influential or central nodes shift collective preferences in larger social networks, and how this herding effect is used by major brands to shape consumer trends and drive marketing outcomes.

Base Line VS Social Influence VS Celebrity Centrality

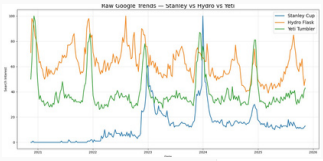
Social Media Driven Herding: The Barbenheimer Phenomenon

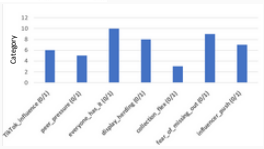




There is a clear example of unexpected herding behaviour: users imitated visible actions (like outfits and double-feature plans), repeated shared jokes, memes, and treated Barbenheimer as something everyone must participate in, creating strong normative herding. This formed a self-reinforcing cycle, where each new wave of posts strengthened the herd's momentum. The momentum became so widespread across platforms that even criticism and backlash highlighted how socially dominant the trend had become.

Social Influence Herding of the Stanley cup trend

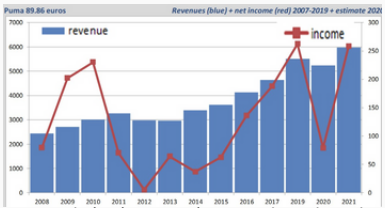




Across all platforms, we find strong evidence of herding behaviour in the adoption of the Stanley Cup tumbler. Search interest, social media visibility, and community discussions all show that individuals are influenced by their peers. Display herding (show-off) is the most common behaviour, followed by informational herding (search spikes) and normative pressure ("everyone has it").

The Stanley marketing adopted a decentralised, peer-driven social influence rather than top-down marketing or celebrity promotion. This aligns with known patterns of viral consumer behaviour observed through research, and works better for the nature of the product.

Puma's Rise Through Celebrity Centrality and Herding Behavior



Puma's evolution into a mainstream brand was heavily reliant on their celebrity collaboration. The strategic use of Rihanna's influence, combined with her genuine creative involvement, created strong brand loyalty and increased their woman-centric market segment, resulting in increased sales and a larger market share.

Puma's motorsport collaborations use top racers (high-centrality figures) to attract the male audience. This fuels a cycle where celebrities endorse a brand, making it a new culture everyone wants to be a part of. By tying the idea of "winners" to wearing Puma and releasing Limited Edition drops that trigger urgency and FOMO, the brand amplifies both social proof and crowd behaviour and strengthens its identity and drives long-term revenue growth.

Conclusion:

Across the discussions and real-world cases, outcomes were shaped by central individuals and collective herding. People aligned with strong, visible voices or converged simply by following each other. This shows that influence can flow from central nodes or emerge organically from crowd behavior, displaying how easily opinions shift through both influence and imitation.